

MediaTek Inc.

Addressing Key Investor Questions

Overweight

2454.TW, 2454.TT

Price (01 Jul 26): NT\$4,335.00

Price Target (Jun-27): NT\$5,300.00

1. **TPU v8t (Zebrafish) ramping up on track, with potential for upside to 2027 revenue guidance:** We believe TPU v8t is on track for a volume ramp-up in 4Q26, supporting the company's target of US\$2bn in ASIC revenue (or 400-500k unit shipments) this year. For 2027, management has guided to US\$7-12bn in ASIC revenue (implying 10-15% share of a US\$70-80bn ASIC market), underpinned by strong customer demand and confidence in material availability. We have seen Google TPU demand skewing toward our previous bull-case scenario and have revised up our TPU estimates in May (now forecasting 8mn market TPU shipments in 2027 vs. 5mn units expected six months ago ([latest CoWoS report](#))). However, we still see upside to our current MediaTek enterprise ASIC revenue estimate of US\$10.8bn in 2027 (~2.5mn TPU v8t units, ~30% share of 8mn total TPU shipments), with further upside beyond US\$12bn achievable if MediaTek can secure additional substrate and 3nm wafer capacity.
2. **TPU v9 Humufish design status on track to tape out in the next 2-3 quarters, market share likely to skew to the upside, given challenges with competitor program:** We believe TPU v9 (Humufish) is on track to tape out in late 2026-early 2027, with revenue contribution starting in late 2027 or early 2028. We believe that v9 Humufish ASP should be 2-3x that of v8t (US\$12k+ vs. ~US\$4.3k), reflecting significantly greater design complexity (4 N2 compute dies, 4 I/O dies, and 4 memory I/O dies vs. 1 N3 compute die and 1 I/O die in v8t). Meanwhile, our supply chain checks continue to indicate that the key competing project (Pumafish) from US Fabless competitor has been either canceled or postponed, creating an opportunity for MediaTek to gain share in 2028-29. While we do not expect the competitor to be entirely absent - likely launching Sunfish v8i variant with 4 N3 based compute dies - we still expect MediaTek to benefit from an earlier mass-production timeline (tape-out at least 1-2 quarters ahead of peers' v9). We are currently expecting ~2M units for Humufish v9 units in 2028, but do expect further upside, depending on Intel EMIB-T packaging yield progress. Our recent research indicates an extension of v9 Humufish (called Triggerfish), with specific design changes (including more SRAM and potentially a controller die within the package) to extend Humufish to accommodate inference needs as well. This could also result in upside to Mediatek's TPU v9 family shipment expectations in 2028-29.
3. **TPUv10 RFQ in progress, engagement model similar to v9, confident of securing a key project:** MediaTek is currently engaged in the TPU v10 RFQ process, based on our checks, with results expected in the coming months. We see a high probability of MediaTek securing at least one TPU v10 (Icefish) project, with Broadcom supporting another v10 project. While there is some discussion around additional v10 variants—potentially including a full CoT model—our checks indicate that GOOG is still embracing a semi-COT model for the v10 family. Key IP that Mediatek is providing in the v10 generation would be 448 G High Speed electrical SERDES, die-to-die interconnect and experience in 3D packaging – we still do not see Google embracing CPO for its TPU projects in the v10 generation. We do see the possibility of further accelerator variants in the v9 family, including a fast-inference chip likely

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awarded to Marvell (SRAM based inference, codenamed Merope), as Google's Compute program expands in scope and attempts to match workloads with specific ASICs.

4. **More progress on new customer RFQs; increased rack-level design from smaller customers; Tesla most likely in the next 1-2 quarters:** We believe MediaTek is in RFQ discussions with several hyperscalers (we believe Tesla, Meta, and Microsoft, etc.). While all are still in RFQ stage, we see the best chance of a win at Tesla, given its need for more front-end design support, which aligns with MediaTek's strengths. While any win would not contribute revenue before late 2028-2029, confirmation would be positive for sentiment as it demonstrates diversification away from a single large customer. We believe that MediaTek is now securing multiple RFQs from various ASIC design service projects, after the successful tape out of TPUv8t. Several of these projects are also seeing some design engagement at a rack-level basis, rather than just for ASIC chip design.
5. **Intel EMIB-T likely for TPUv9; a return to TSMC 3D SoIC is possible for TPU v10:** We now believe that MediaTek has finalized Intel EMIB-T as the primary packaging choice for TPU v9 Humufish, with CoWoS based solution unlikely to be pursued. Based on our checks, EMIB yield for large-size packaging remains low (likely around 60%), but MediaTek appears to be confident that yield can improve before mass production ramps up in late 2027 and the chance of meaningful execution risk remains low. However, for TPU v10, we believe Google is considering a 3D packaged IC solution for logic-logic integration, which may suggest a return to TSMC SoIC packaging, along with CoWoS-L. We believe that Intel's 3D packaging (Foveros Direct) is still at a very early stage of ramp as compared to TSMC SoIC. For logic-HBM integration, we also believe that TSMC is now aggressively pushing for larger reticle size CoWoS-L / CoPoS in the 2029-30 timeframe, which will align with the mass production ramp timeline for TPU v10.
6. **Concerns about competition from QCOM in Datacenter ASIC are early and overblown:** QCOM (covered by Joseph Cardoso) announced last week at its investor day that it is expanding into the DC field, partnering with Meta (on C1000 datacenter CPU) and Microsoft (on its AI inference chip using HBC memory). Meanwhile, QCOM highlighted two custom silicon hyperscaler clients (one likely ByteDance, the other likely AWS, in our view) targeting US\$1+bn revenue each in 2027. We agree that the recent Alphawave acquisition gives QCOM access to SerDes IP, and its chip-design pedigree is strong given the long experience in smartphone/CE. However, prior experience in smaller consumer CPUs may take time to translate into a reliable track record in large AI accelerators. MediaTek, in comparison, has been engaged in TPU projects since 2020 and has been developing its Serdes IP since 2018, before gaining firm traction. QCOM has two advantages at this point, in our view: (1) Architectural license access for ARM instruction set, which can help it differentiate in ARM CPUs over time, and (2) Strong N3 wafer allocation at TSMC, which, in the current shortage environment, could translate into datacenter ASIC revenues, especially from customers like AWS, even if design engagement is not very high. Based on our checks, we do not see any large AI ASIC projects based on CoWoS or 2.5D packaging, where QCOM is the lead design service partner.
7. **Gross margins to dip modestly, while OP margins should ramp up meaningfully as DC ASIC business ramps up:** We expect MediaTek GM to dip by 2-3ppts YoY in 2026 to 45-46%, reflecting ASIC dilution and a smaller smartphone revenue base amid weak end demand. Rising input costs (wafers, OSAT, substrates), on the other hand, should be broadly offset by the late-2026 price hikes, keeping margins stable throughout 2026. While ASIC GM is below the corporate average, its OPM contribution is accretive. As DC ASIC business ramps, we expect GMs to settle at ~42% in 2028, while OP margins could move up to ~25%. We model the datacenter ASIC business to represent ~50%/~70% of company OP in 2027/28, helping expand OPM to ~20%/~25% (vs. 17% in 2025).
8. **A long-term pursuit of full-COT from GOOG is unavoidable, but will take time:** A

key question on the durability of MediaTek's ASIC business is GOOG's pursuit of a full-CoT (Customer Owned Tooling) business model, similar to AWS for Trainium. In the very long term, we do believe GOOG will attempt to move towards a full CoT model to take more control of the chip design, but for designs until TPU v10 we are not seeing any change from the current semi-CoT approach. In the next 3-4 years, we believe that MediaTek is likely to benefit from the rising adoption of this semi-CoT model, and should gain meaningful share against Broadcom, given the 40-50% cheaper offering. At the same time, smaller projects within GOOG's ASIC portfolio (such as Server CPU, video accelerators) could migrate to a full-CoT model much more quickly, and more design service vendors are also entering the supply chain, such as Marvell and Alchip. We believe that MediatTek is fully cognizant of this longer-term tail risk and is cultivating other ASIC projects to mitigate exposure, as well as move into rack-level design to enhance its value add to new ASIC customers.

9. **Investor feedback – increased interest from Global Tech investors, next watch point is second customer confirmation:** We have seen a much broader set of global tech investors express interest in MediaTek stock in the last two months, after the successful tapeout of its TPUv8t project for GOOG. If MediaTek is able to move TPUv9 to tapeout on time, we expect global tech investor interest to surge meaningfully, given its position as a credible competitor to Broadcom. For Asian investors, TPU expectations have already moved up meaningfully and the next watch point would be the confirmation of a second ASIC customer (we believe Tesla is quite likely), and potential details of project award for TPU v10.
10. **Core business units struggling but broad-based price hikes to support revenues and gross margins:** Unit demand in core businesses (smartphones, TVs, and consumer IoT) continues to be challenging due to weaker consumer demand and impact from rising memory prices, but MediaTek is still able to see healthy growth in computing, 5G thin modems, and automotive. In addition, MediaTek is raising prices across its core business, averaging 5-10%, on our estimates, with the increases flowing through revenue in late 3Q26 to early 2027. This should stabilize GMs in core businesses, as the increases are largely intended to offset rising costs from OSATs, substrates, and foundry price adjustments. We expect non-ASIC revenue to grow mildly in 2026, despite challenges in consumer demand.

Investment Thesis, Valuation and Risks

MediaTek Inc. *(Overweight; Price Target: NT\$5,300.00)*

Investment Thesis

We are Overweight on MediaTek, given (1) MediaTek's v9 TPU project (Humufish) appearing to emerge as the primary v9 family in 2028-29, (2) MediaTek's smoother execution with v9 positioning it well to maintain its leadership in v10 competition, and (3) solid prospects in datacenter solutions, ranging from XPU's to interconnect solutions, custom HBM and full-rack-level enablement, with strong engagement for new XPU projects from most hyperscalers. Key catalysts from here on, in our view, are (1) indications of a larger role with TPU v9 if EMIB-T packaging execution goes on track, (2) the announcement of new customers before the end of the year – we believe Meta, Tesla are key potential customers, (3) the expansion of the datacenter TAM to include interconnect solutions and rack-level implementation beyond the top 4 CSPs.

Valuation

Our Jun-27 PT of NT\$5,300 is based on our SoTP valuation (consisting of 12x 12M fwd core earnings + 40x 12M fwd ASIC-related earnings).

Risks to Rating and Price Target

Key downside risks include: (1) further pressure on Smartphone SoC GMs due to cost escalations and increased competition; and (2) slower progress on the NVDA partnership, given NVDA's expanded engagement with Intel and other chip makers.

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MediaTek Inc. (2454.TW, 2454 TT) Price Chart



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30-Jul-23	N	658.00	730
17-Oct-23	N	840.00	780
29-Oct-23	N	801.00	750
05-Jan-24	N	928.00	800
01-Feb-24	N	966.00	880
11-Apr-24	N	1195.00	1,050
28-Apr-24	N	1005.00	1,060
29-Jul-24	N	1245.00	1,050
01-Aug-24	N	1220.00	1,150
22-Oct-24	N	1330.00	1,175
31-Oct-24	N	1290.00	1,180
05-Feb-25	N	1490.00	1,275
27-Apr-25	N	1380.00	1,350
16-Jun-25	N	1250.00	1,300
25-Nov-25	N	1150.00	1,260
28-Jan-26	N	1775.00	1,525
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31-May-26	OW	4310.00	5,300

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